# Diego De Pablo



## Software Engineer | Backend, Data & Applied AI

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## Professional Summary

Software engineer and bioinformatics graduate focused on backend development, data engineering, and applied AI. Experience building modular web services, concurrent data processing pipelines, and semantic search systems with model integration. Graduated with academic honours and currently pursuing an M.S. in Computer Engineering.

## Professional Experience

**Software Engineer, Khaos Research** — **Sept 2025 – Present**

- Engineered backend services and integration modules for the European **SEDIA** project, implementing federated and interoperable data space specifications under Gaia-X standards.
- Architected and integrated a hybrid search engine combining dense vector retrieval in **Qdrant** with BM25 lexical search in **Elasticsearch**, structured via schema-validated **Pydantic AI** agents.
- Built the concurrent **MetaDataXtract** pipeline for scientific dataset ingestion and metadata extraction across +30 formats, featuring SHA-256 integrity verification and automated file security analysis via the VirusTotal API.

**Full-Stack Developer, mSurgery** — **March 2025 – July 2025**

- Migrated core frontend modules to **Svelte 5** with runes, decoupling state management to optimize UI reactivity with the 3D surgical viewport.
- Developed interactive interfaces for the **SMOTTS** surgical simulation platform, integrating real-time digital twin telemetry for simulated procedures.
- Delivered technical platform demonstrations in English to international investors and evaluation panels at the National Digital Content Hub (Polo Digital).

**Junior Software Engineer, T&B Company Group** — **June 2021 – Dec 2021**

- Refactored legacy modules in **Java** and **Angular** into decoupled architectures, resolving production bugs and improving codebase maintainability.
- Authored standardized API contracts with **OpenAPI/Swagger** and documented REST microservices to streamline cross-team service integration.

## Selected Technical Projects

### MetaDataSearch: Semantic Search Engine

**Stack:** Next.js, FastAPI, Qdrant, Elasticsearch, Pydantic AI, Docker

- Hybrid search platform for scientific data catalogs: combines dense embeddings and BM25 lexical retrieval, generating structured semantic filters from natural language queries.

### Biomedical Remote Monitoring IoT (Bachelor's Thesis)

**Stack:** C++, FreeRTOS, ESP32, MAX30102, MPU-6050, MQTT, FastAPI, TimescaleDB

- End-to-end telemetry prototype: ESP32 firmware for vital sign acquisition via PPG (heart rate and SpO2) and fall detection via accelerometry, synchronizing over MQTT with a web dashboard. **Graded 10/10 with Highest Honours**.

### TITAN Workflow: AI Scientific Orchestration

**Stack:** FastAPI, Pydantic AI, LangChain, Docker, LLM

- AI-assisted workflow composition service: validates data dependencies and I/O schemas using strict Pydantic AI contracts prior to execution graph scheduling.

## Education

**B.S. in Health Engineering (Bioinformatics Specialization)** | *University of Málaga* — **2021 – 2025**

- Matrícula de Honor (Highest Honours) in core coursework: **Programming Fundamentals**, **Intelligent Systems**, and **Bioinformatics Projects**.
- Bachelor's Thesis in Biomedical IoT graded **10/10 with Matrícula de Honor (Highest Honours)**.

**M.S. in Computer Engineering** | *University of Málaga* — **2025 – Present**

- Advanced coursework in distributed systems, scalable data architectures, and applied machine learning.

## Technical Skills & Languages

**Languages:** Python, TypeScript, JavaScript, C++, SQL, Java, Bash, R.

**Backend & Data:** FastAPI, PostgreSQL, Elasticsearch, TimescaleDB, Qdrant (Vector DB), REST APIs, OpenAPI/Swagger.

**AI & Machine Learning:** Pydantic AI, LangChain, LLMs / Embeddings, PyTorch (academic & project-level).

**DevOps, Systems & Web:** Docker, Kubernetes, Argo CD, CI/CD (GitHub Actions), Linux, FreeRTOS (ESP32), React, Svelte 5.

**Languages: Spanish** (Native) · **English** (B2 Professional / full working proficiency, international technical demos).